

## 第一章 解 $AX=b$ 的直接法

§1.1 三角方程组  $\rightarrow$  前代法, 后代法

§1.2 高斯消元法,  $A=LU$

§1.3 选主元消元法,  $\begin{cases} \text{全主元消元法} & PAQ=LU \\ \text{列主元消元法} & PA=LU \end{cases}$

§1.4 直接分解法

(一) 对称正定矩阵的平方根法

$A \in \mathbb{R}^{n \times n}$  是对称正定的, 即  $A$  满足  $A=A^T$ , 且  $x^T A x > 0$ ,  $x \in \mathbb{R}^n$ ,  $x \neq 0$ .

(I) 定理: 若  $A$  对称正定, 则存在一个对元均为正数的下三角阵  $L$ , 使得  $A=LL^T$ .

证明:  $A$  正定, 则  $A$  的全部顺序主子阵均正定,  $\rightarrow$  可逆

由高斯消元法的LU分解定理:

存在一个单位下三角阵  $\tilde{L}$  和一个上三角阵  $U$ , 使得  $A=\tilde{L}U$

$\because A$  正定  $\Rightarrow \det A \neq 0 \Rightarrow \det \tilde{L} \neq 0$ ,  $\det U = u_{11}u_{22}\cdots u_{nn} \neq 0$

$\Rightarrow u_{ii} \neq 0, i=1, \dots, n$

令  $D = \text{diag}(u_{11}, u_{22}, \dots, u_{nn})$  则  $D$  可逆

记  $\tilde{U} = D^{-1}U$ , 则  $\tilde{U}$  为单位上三角阵.

且  $A = \tilde{L}D\tilde{U}$   $A^T = \tilde{U}^T D \tilde{L}^T$

$$\begin{pmatrix} \frac{1}{u_{11}} & & \\ & \ddots & \\ & & \frac{1}{u_{nn}} \end{pmatrix} \begin{pmatrix} 1 & & \\ * & \ddots & \\ & & 1 \end{pmatrix} \begin{pmatrix} u_{11} & & \\ & \ddots & \\ & & u_{nn} \end{pmatrix}$$

$$\text{由 } A=A^T \Rightarrow \tilde{L} D \tilde{U} = \tilde{U}^T D \tilde{L}^T \Rightarrow D^T \tilde{U}^{-T} \tilde{L} D = \tilde{L}^T \tilde{U}^{-1}$$

$$\Rightarrow D^T \tilde{U}^{-T} \tilde{L} D = \tilde{L}^T \tilde{U}^{-1} = I$$

$$\Rightarrow \tilde{U} = \tilde{L}^T \Rightarrow A = \tilde{L} D \tilde{L}^T$$

另一个  $D$  的对角元为正, ( $x \neq 0, x^T A x > 0 \Rightarrow x^T \tilde{L} D \tilde{L}^T x > 0$ )

$$\Rightarrow (\tilde{L}^T x)^T D (\tilde{L}^T x) > 0$$

$\Rightarrow D^{\frac{1}{2}} = \text{diag}(\sqrt{a_{11}}, \dots, \sqrt{a_{nn}}) \because x \text{ 任意, } \tilde{L} \text{ 可逆} \Rightarrow D > 0$

令  $L = \tilde{L} D^{\frac{1}{2}}$ , 则  $L$  的对角元为正.

$$A = L L^T. \quad \text{证.}$$

注: 此定理为平方根分解定理, 又称为 Cholesky 定理,  $L$  为 Cholesky 因子

(II) 如何求  $L$ ?

$$\begin{pmatrix} a_{11} & a_{12} & \dots & a_{1n} \\ a_{21} & a_{22} & \dots & a_{2n} \\ \vdots & \vdots & \ddots & \vdots \\ a_{n1} & a_{n2} & \dots & a_{nn} \end{pmatrix} = A = L L^T = \begin{pmatrix} l_{11} & l_{21} & \dots & l_{n1} \\ 0 & l_{22} & \dots & 0 \\ \vdots & \vdots & \ddots & \vdots \\ 0 & 0 & \dots & l_{nn} \end{pmatrix} \begin{pmatrix} l_{11} & l_{21} & \dots & l_{n1} \\ 0 & l_{22} & \dots & 0 \\ \vdots & \vdots & \ddots & \vdots \\ 0 & 0 & \dots & l_{nn} \end{pmatrix}$$

对比  $A = L L^T$  两边元素有.

第一列:  $a_{11} = l_{11}^2 \Rightarrow l_{11} = \sqrt{a_{11}} \quad l_{i1} = a_{i1} / l_{11}$

$$a_{i1} = l_{i1} l_{11}$$

若已计算出  $L$  的前  $k-1$  列元素, 比较第  $k$  列元素有

$$a_{kk} = \sum_{p=1}^k l_{kp} l_{kp} = \sum_{p=1}^k l_{kp}^2 = \sum_{p=1}^{k-1} l_{kp}^2 + l_{kk}^2 \Rightarrow l_{kk} = \sqrt{a_{kk} - \sum_{p=1}^{k-1} l_{kp}^2}$$

$$a_{ik} = \sum_{p=1}^k l_{ip} l_{kp} = \sum_{p=1}^{k-1} l_{ip} l_{kp} + l_{ik} l_{kk} \quad i=k+1, \dots, n$$

$$\Rightarrow \begin{cases} l_{kk} = \sqrt{a_{kk} - \sum_{p=1}^{k-1} l_{kp}^2} & 10^{-16} \quad 10^{-8} \\ l_{ik} = (a_{ik} - \sum_{p=1}^{k-1} l_{ip} l_{kp}) / l_{kk} \end{cases} \quad i=k+1, \dots, n$$

对  $k=1, 2, \dots, n$  都适用

**算法 1.3.1** 运算量  $O(n^3)$ . 为 Gauss 消元法的一半.

若  $AX=b$  且系数矩阵  $A>0$ , 求解  $AX=b$  的步骤:

① 用算法 1.3.1 计算  $A$  的平方根分解  $A=LL^T$ .

② 用算法 1.1.1 计算  $Ly=b$  得  $y$

③ 用算法 1.1.2 计算  $L^T x=y$ . 得  $x$ .

(二) 改进的平方根法 (正定阵的 LDL<sup>T</sup>分解) (为了避免开根号)

同 (一) 的定理证明: 存在单位下三角阵  $L$  和一个对角阵  $D$ , 使得  $A=LDL^T$ .

$$\begin{pmatrix} a_{11} & a_{12} & \dots & a_{1n} \\ a_{21} & a_{22} & \dots & a_{2n} \\ \vdots & \vdots & \ddots & \vdots \\ a_{n1} & a_{n2} & \dots & a_{nn} \end{pmatrix} = \begin{pmatrix} 1 & & & \\ l_{21} & \ddots & & \\ \vdots & \ddots & \ddots & \\ l_{n1} & \dots & l_{nm} & 1 \end{pmatrix} \begin{pmatrix} d_1 & & & \\ & \ddots & & \\ & & d_n & \\ & & & \ddots \end{pmatrix} \begin{pmatrix} 1 & l_{12} & \dots & l_{1n} \\ & 1 & & \\ & & \ddots & \\ & & & 1 \end{pmatrix}$$

$$= \begin{pmatrix} 1 & & & \\ l_{21} & \ddots & & \\ \vdots & \ddots & \ddots & \\ l_{n1} & \dots & l_{nm} & 1 \end{pmatrix} \begin{pmatrix} d_1 & d_1 l_{21} & \dots & d_1 l_{1n} \\ & d_2 & \dots & d_2 l_{n2} \\ & & \ddots & \\ & & & d_n \end{pmatrix}$$

对比两边元素,

第 1 列:  $a_{11} = d_1$   
 $a_{i1} = l_{i1} d_1 \Rightarrow \begin{cases} d_1 = a_{11} \\ l_{i1} = a_{i1} / d_1 \end{cases} \quad i=2, \dots, n$

若已计算出  $d_1, \dots, d_{k-1}$ ,  $L$  前  $k-1$  列元素, 对比第  $k$  列有.

$i=k, \dots, n$

$$a_{ik} = \sum_{p=1}^n l_{ip} b_{pk} = \sum_{p=1}^n l_{ip} d_{pk} = \sum_{p=1}^{k-1} l_{ip} d_{pk} + l_{ik} d_k \quad (l_{kk}=1)$$

$$\Rightarrow d_k = a_{kk} - \sum_{p=1}^{k-1} d_p l_{kp}^2$$

$$l_{ik} = (a_{ik} - \sum_{p=1}^{k-1} l_{ip} d_{pk}) / d_k, \quad i=k+1, \dots, n$$

对  $k=1, 2, \dots, n$  都适用

**算法 1.3.2** 运算量  $O(n^3)$

求解  $Ax=b$  ( $A>0$ ) 的步骤:

① 用算法 1.3.2 算出  $A = LDL^T$

② 用算法 1.1.1 计算  $Ly=b$ , 得  $y$ .

③ 计算  $Dz=y \Rightarrow z$ .

④ 用算法 1.1.2 计算  $L^T x = z$ , 得  $x$

输入:  $A$

输出:  $D, L \rightarrow$  存在  $A = LDL^T$  因子

$$Ax=b \Leftrightarrow \underbrace{LDL^T}_{\substack{y \\ z}} x = b$$

例: 利用改进的平方根法求  $Ax=b$ . 其中

$$A = \begin{pmatrix} 4 & -2 & 4 \\ -2 & 10 & -2 \\ 4 & -2 & 8 \end{pmatrix} \quad b = \begin{pmatrix} 4 \\ 16 \\ 8 \end{pmatrix}$$

$$\text{解: } A = \begin{pmatrix} 1 & & \\ l_{21} & 1 & \\ l_{31} & l_{32} & 1 \end{pmatrix} \begin{pmatrix} d_1 & & \\ & d_2 & \\ & & d_3 \end{pmatrix} \begin{pmatrix} 1 & l_{21} & l_{31} \\ & 1 & l_{32} \\ & & 1 \end{pmatrix}$$

$$= \begin{pmatrix} 1 & & \\ l_{21} & 1 & \\ l_{31} & l_{32} & 1 \end{pmatrix} \begin{pmatrix} d_1 & d_1 l_{21} & d_1 l_{31} \\ & d_2 & d_2 l_{32} \\ & & d_3 \end{pmatrix}$$



$$\Rightarrow d_1 = a_{11} = 4$$

$$a_{21} = -2 = l_{21} d_1 \Rightarrow l_{21} = -\frac{1}{2} \quad a_{31} = 4 = l_{31} d_1 \Rightarrow l_{31} = 1$$

$$a_{22} = d_1 l_{21}^2 + d_2 \Rightarrow d_2 = 10 - 4 \cdot \left(-\frac{1}{2}\right)^2 = 9$$

$$a_{32} = l_{31} d_1 l_{21} + l_{32} d_2 \Rightarrow l_{32} = (-2 - 1 \cdot 4 \cdot (-\frac{1}{2})) / 9 = 0$$

$$a_{33} = d_1 l_{31}^2 + d_2 l_{32}^2 + d_3 \Rightarrow d_3 = 8 - 4 \cdot 1^2 - 9 \cdot 0^2 = 4.$$

$$L = \begin{pmatrix} 1 & 0 & 0 \\ -\frac{1}{2} & 1 & 0 \\ 1 & 0 & 1 \end{pmatrix} \quad D = \begin{pmatrix} 4 & & \\ & 9 & \\ & & 4 \end{pmatrix} \quad L^T = \begin{pmatrix} 1 & -\frac{1}{2} & 1 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{pmatrix}$$

$$Ax=b \quad 1^\circ \quad Ly=b = \begin{pmatrix} 4 \\ 16 \\ 8 \end{pmatrix} \quad y = (y_1, y_2, y_3)^T$$

$$y_1 = 4, \quad -\frac{1}{2}y_1 + y_2 = 16 \Rightarrow y_2 = 18$$

$$y_1 + y_3 = 8 \Rightarrow y_3 = 4.$$

$$y = \begin{pmatrix} 4 \\ 18 \\ 4 \end{pmatrix}$$

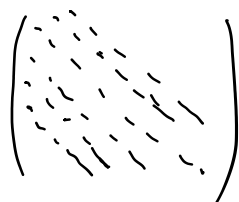
$$2^\circ \quad Dz=y \Rightarrow z = (1, 2, 1)^T$$

$$3^\circ \quad L^T x = z \Rightarrow x_3 = 1 \quad x_2 = 2 \quad x = \begin{pmatrix} 1 \\ 2 \\ 1 \end{pmatrix}.$$

$$x_1 - \frac{1}{2}x_2 + x_3 = 1 \Rightarrow x_1 = 1$$

□

### (三) 带状矩阵分解

(I)  非零元素分布在对称线元素的两侧。  
若上侧有  $m_1$  条及对称线上有非零元素，下侧有  $m_2$  条  
及对称线上有非零元素，称带宽为  $m_1 + m_2 + 1$  的带状矩阵。

若其各阶顺序主子式不为 0，则可分解为

带宽为  $m_1 + 1$  的下三角阵与带宽为  $m_2 + 1$  的上三角阵的乘积。(思考)

(II) 三对角线矩阵

例.  $A = \begin{pmatrix} a_1 & b_1 & & \\ c_1 & a_2 & b_2 & \\ & c_2 & a_3 & b_3 \\ & & \ddots & \ddots \\ & & & c_{n-1} & a_n & b_n \end{pmatrix}$  →  $m_1 = m_n = 1$

$$= \underbrace{\begin{pmatrix} \alpha_1 & & & \\ r_1 & \alpha_2 & & \\ 0 & r_2 & \alpha_3 & \\ & \ddots & \ddots & \ddots \\ & & r_{n-1} & \alpha_n \end{pmatrix}}_L \underbrace{\begin{pmatrix} 1 & \beta_1 & & \\ & 1 & \beta_2 & \\ & & \ddots & \ddots \\ & & & 1 & \beta_{n-1} \\ & & & & 1 \end{pmatrix}}_U$$

$l_{ii} = \alpha_i, \quad l_{i,i-1} = r_i, \quad u_{ii} = 1, \quad u_{i,i+1} = \beta_i$

$$\Rightarrow c_i = a_{i,i-1} = \sum_{p=1}^n l_{ip} u_{p,i-1} = l_{i,i-1} u_{i-1,i-1} + l_{ii} \underbrace{u_{i,i-1}}_{=0} = r_{i-1}$$

$$a_i = \sum_{p=1}^n l_{ip} u_{p,i} = l_{i,i-1} u_{i-1,i} + l_{ii} u_{ii} = r_i \beta_{i-1} + \alpha_i$$

$$b_i = a_{i,i+1} = \sum_{p=1}^n l_{ip} u_{p,i+1} = l_{i,i-1} \underbrace{u_{i-1,i+1}}_{=0} + l_{ii} u_{i,i+1} = \alpha_i \beta_i$$

$$\Rightarrow \begin{cases} r_i = c_i \\ \alpha_i = a_i - r_i \beta_{i-1} \\ \beta_i = b_i / \alpha_i \end{cases}$$

(注)  $\beta_0 = 0$

$i = 1, \dots, n$

追赶法: 求  $Ax = f = (f_1, f_2, \dots, f_n)^T$ ,  $A$  为三对角线阵

$$Ly = f \Rightarrow y_i = (f_i - c_i y_{i-1}) / \alpha_i, \quad i = 1, \dots, n$$

$$Ux = y \Rightarrow x_i = y_i - \beta_i x_{i+1}, \quad i = n, n-1, \dots, 1$$

(注)  $\beta_n = 0$

$$\begin{cases} \alpha_i = a_i - c_i \beta_{i-1} \\ \beta_i = b_i / \alpha_i \\ y_i = (f_i - c_i y_{i-1}) / \alpha_i \end{cases} \quad i = 1, \dots, n$$

$$x_k = y_k - \beta_k x_{k+1}, \quad k = n, n-1, \dots, 1$$

(四) 一般矩阵 (满足高斯消元法条件的实矩阵)

(I) Doolittle 分解:  $A = L U$  (行和列)  
 单位下  $\swarrow$   $\searrow$  上三角

(II) Courant 分解  $A = L U$   
 下三角  $\swarrow$   $\searrow$  单位上

(五) 扩展

(I) 其他方法: 参数法, 特征值法

(II) Gauss-Jordan 消元  $AB = I$   $B = (b_1 \dots b_n)$   $Ab_i = e_i$

照抄  $(A \ I) \rightarrow (I \ B)$   $PA = I$   $P = B = A^{-1}$

$$\begin{pmatrix} a_{11} & a_{12} & \dots & a_{1n} & & \\ a_{21} & a_{22} & \dots & a_{2n} & & \\ \vdots & \vdots & & \vdots & & \\ a_{n1} & \dots & \dots & a_{nn} & & \end{pmatrix} \xrightarrow{\text{Gauss消元}} \begin{pmatrix} a_{11} & a_{12} & \dots & a_{1n} & 1 & 0 & \dots & 0 \\ 0 & \tilde{a}_{22} & \dots & \tilde{a}_{2n} & * & 1 & & \\ \vdots & \vdots & & \vdots & \vdots & \vdots & & \\ 0 & \tilde{a}_{n2} & \dots & \tilde{a}_{nn} & * & & & 1 \end{pmatrix}$$

$$\xrightarrow{\text{第1行} \times 1/a_{11}} \begin{pmatrix} 1 & \tilde{a}_{12} & \dots & \tilde{a}_{1n} & 1/a_{11} & 0 & \dots & 0 \\ 0 & \tilde{a}_{22} & \dots & \tilde{a}_{2n} & * & 1 & & \\ \vdots & \vdots & & \vdots & \vdots & \vdots & & \\ 0 & \tilde{a}_{n2} & \dots & \tilde{a}_{nn} & * & & & \end{pmatrix}$$

$$\xrightarrow{\begin{matrix} \text{第2行} \times * \text{倍加到} \\ \text{第1, 3, 4, } \dots \text{ n行.} \end{matrix}} \begin{pmatrix} 1 & 0 & * & * & \dots & * & \vdots & * & * \\ 0 & \tilde{a}_{22} & * & \dots & * & \vdots & * & * & \\ 0 & 0 & * & & \vdots & \vdots & * & * & 1 \\ \vdots & \vdots & \vdots & & \vdots & \vdots & \vdots & \vdots & \vdots \\ 0 & 0 & * & \dots & * & \vdots & * & * & \end{pmatrix}$$

$$\xrightarrow{\text{第2行} \times 1/\tilde{a}_{22}} \begin{pmatrix} 1 & 0 & * & \dots & * & \vdots & * & * \\ 0 & 1 & * & \dots & * & \vdots & * & * \\ 0 & 0 & * & \dots & * & \vdots & * & * \\ \vdots & \vdots & \vdots & & \vdots & \vdots & \vdots & \vdots \\ 0 & 0 & * & \dots & * & \vdots & * & * \end{pmatrix}$$